

Niek Kamer

Perf engineering on zk proof systems. Plonky3 maintainer.

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Available for zk perf engineering contracts and senior IC roles.

SELECTED OPEN-SOURCE WORK

Plonky3 — maintainer (May 2026 – present)

github.com/Plonky3/Plonky3

- [PR #1623](#) (merged): `add_canonical_asm` skips redundant canonicalize on provably-canonical sites. -2.93% end-to-end Goldilocks Poseidon2 prove time on aarch64 NEON; -5.91% / -6.21% on external permute kernels.
- [Issue #1642](#) (audit) → [PR #1645](#) by @Nashtare: packed AVX-2 Poseidon1 running 0.77× scalar due to missing Karatsuba back-port. 1.20× wall-time speedup post-fix.
- [PR #1656](#) (in review): packed-NEON Poseidon2 dispatch routed through generic. -16.4% Merkle wall time on Pi 5 Cortex-A76 across $\log_2$ leaves $\in \{14, 16, 18\}$.
- [PR #1667](#) (in review): cross-permute batched Poseidon2 for widths 8 and 12. -27.2% Zen 5 AVX-2, -31.5% Zen 5 AVX-512, -10.82% Pi 5 NEON (per-permute, width 8).

zkmcu — author

github.com/Niek-Kamer/zkmcu

- `no_std` Rust Groth16/BN254 verifier for microcontrollers. Cortex-M33 and RISC-V Hazard3 targets. ~1 s verify on a \$7 RP2350.
- Pairing arithmetic, optimized BN254 G1/G2 ops, fits within RP2350 RAM budget.

poseidon2-harness — author

github.com/Niek-Kamer/poseidon2-harness

- Dual-tree differential testing + PMU-cycle benches for Plonky3 Poseidon2 on aarch64 NEON.
- Pulls two Plonky3 commits side-by-side via cargo git deps. Bitwise + canonical oracles across both trees, per-layer + full-permute coverage, widths 8 / 12 / 16 / 20.
- Used during a Plonky3 Poseidon2 audit; turned up two correctness-adjacent findings.

SELECTED WRITING

All at niek-kamer.github.io/writing. Methodology + measurement write-ups documenting the contributions above.

- The trait-method boundary that cost Plonky3 Poseidon2 27% (2026-05-17)
- The AVX-512 kernel that was 1.79× slower than the compiler's (2026-05-16)
- The Poseidon2 regression my microbenchmark told me wasn't there (2026-05-15)
- The SIMD path that was 0.77× scalar (2026-05-13)
- The two instructions hiding in every Goldilocks Poseidon2 add (2026-05-13)

SKILLS

Languages. Rust (primary). TypeScript, C, Solidity (reading).

Architectures. aarch64 NEON (Cortex-A76, Apple Silicon code-path via shared NEON kernels), x86 AVX-2 / AVX-512 (Zen 5), Cortex-M33, RISC-V Hazard3.

Performance tooling. `perf_event_open` with grouped counters, `perf annotate` / `perf stat`, criterion, cargo bench, objdump, GDB. Comfortable reading and writing inline NEON / x86 assembly.

Domains. Zero-knowledge cryptography (Groth16, STARKs, FRI, Poseidon1/2 / Goldilocks / Babybear / BN254), `no_std` embedded Rust, perf-counter analysis on superscalar OoO cores.

EXPERIENCE

Founder, Axobello (2025 – present)

Saint Martin / Amsterdam

Independent software work. Current focus: zero-knowledge proof system performance engineering, embedded ZK verification.

EDUCATION

MBO (Netherlands vocational secondary education). Self-taught in systems programming, cryptography, and zero-knowledge proof systems.